Praktikum Hochfrequenz-Schaltungstechnik WS25/26

Bericht

Name:	Sebastian Grigorevski	**Matrikelnr.:**	35690104
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5.5.1

According to the approximation formula of the rise time of DSOs

$$t_o = \frac{0.35}{BW}$$

the rise time for a 60 MHz (120 MHz) DSO are $t_o = 5.8\,\text{ns}$ ($t_o = 2.9\,\text{ns}$) [1]. Thus the DSO cannot resolve rise times lower than those calculated values. The rise and fall times of the inverter are mach faster at 2.05 ns and 1.90 ns (measured with a 350 MHz DSO). Since the rise and fall times are significantly faster the given DSOs are not able to measure them correctly. This can be confirmed by a measurement with a 60 MHz where the rise time was noted at 5.5 ns and the fall time was 4.5 ns being much higher than the actual value.

5.5.2

The formula for reflection coefficients is stated as

$$\Gamma = \frac{Z_\mathrm{t} - Z_0}{Z_\mathrm{t} + Z_0}$$

where Z_t is the termination impedance and $Z_0 = 50\,\Omega$ is the impedance of the wire. For different termination impedances the theoretical reflection coefficients $\Gamma_\mathrm{theo.}$ are calculated in table **??**.

Table 1: theoretical and measured reflection coefficients at a wire impedance of $Z_0 = 50\,\Omega$

Z_t	$50\,\Omega$	$75\,\Omega$	$93\,\Omega$
$\Gamma_\mathrm{theo.}$	0	0.20	0.30
Γ_measured			

Differences can occur due to losses in the wire since the isolation between inner wire and cylindrical conductor deafens the signal due to its magnetic properties.

5.5.3

The cable lengths were determined from the measured time difference between the incident and the reflected pulse. Since the signal propagates to the cable end and back, the cable length is given by

$$l = \frac{c \cdot k \cdot \Delta t}{2}$$

where c is the speed of light and k is the velocity factor of the cable. The time difference Δt was obtained from the oscilloscope display using the number of divisions and the selected time base.

- Measure the time difference Δt between the incident and reflected pulse on the oscilloscope.
- Convert the oscilloscope reading from divisions to time:

$$\Delta t = N_{\text{Div}} \cdot t_{\text{Div}}$$

- Take measurements for both open and short-circuited cable ends.
- Calculate the mean value to reduce measurement uncertainty:

$$\Delta t_{\text{mean}} = \frac{\Delta t_{\text{open}} + \Delta t_{\text{short}}}{2}$$

- Determine the cable length using the cable's velocity factor k:

Cable type	Δt [ns]	k	calc. Length l [m]	Measured Length l [m]
RG58	38.5	0.66	3.8	3.59
RG188	49.0	0.70	4.5	4.52
SAT (white)	30.5	0.79	3.–	3.59

Table 2: Cable lengths calculated from propagation delay measurements

The results obtained for open and short-circuited cable terminations agree within the measurement uncertainty. Remaining deviations are mainly caused by reading errors, additional cable lengths due to T-connectors (and ikea papaer measuring tapes :).

5.5.4

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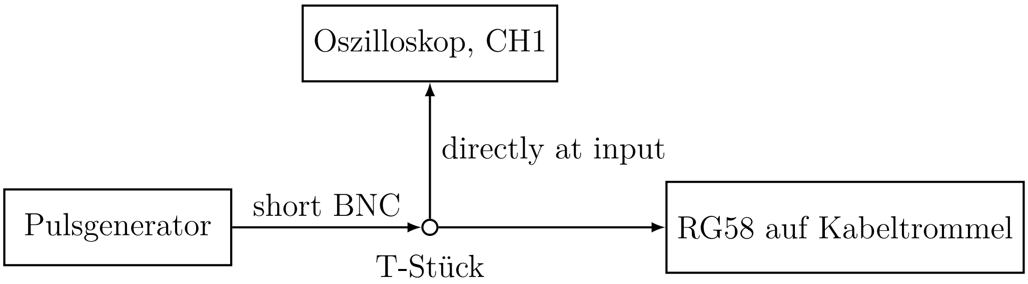

Figure 1: Schaltung zur Messung mit der RG58-Kabeltrommel

5.5.5

5.5.6

three different termination impedances 50, 75 and 93 Ω were connected to the SAT cable to narrow down the impedance range of the cable. Due to the reflections we know that it has to be between 50 and 75 Ω. Using the potentiometer we were able to find the configuration where complete termination occurs and determined the impedance of the SAT cable at 72.8 ±0.2 Ω.

5.5.7

The velocity factor k of the SAT cable can be determined from the known physical length s and the measured round-trip time difference Δt between the incident and reflected pulse. Rearranging the propagation relation

$$v = k\,c = \frac{2s}{\Delta t}$$

gives

$$k = \frac{v}{c} = \frac{2s}{c\,\Delta t}.$$

Inserting the values from Table 2, $s = 3.59\,\mathrm{m}$ and $\Delta t = 30.5\,\mathrm{ns}$, yields

$$k = \frac{2 \cdot 3.59\,\mathrm{m}}{(3.00 \times 10^8\,\mathrm{m/s}) \cdot 30.5 \times 10^{-9}\,\mathrm{s}} \approx 0.79.$$

5.5.8

Assuming that the dielectric of the SAT cable consists of polyethylene, the velocity factor k of an electromagnetic wave in the cable is given by

$$k = \frac{v}{c} = \frac{1}{\sqrt{\varepsilon_r \mu_r}}\,,$$

where c is the speed of light in vacuum, ε_r is the relative permittivity of the dielectric and μ_r its relative permeability. For non-magnetic dielectrics such as polyethylene we can set

$\mu_r \approx 1$, so that

$$k \approx \frac{1}{\sqrt{\varepsilon_r}}\,.$$

Using a typical value of $\varepsilon_r \approx 2.3$ for *solid* polyethylene, the theoretical velocity factor becomes

$$k_{\text{solid PE}} = \frac{1}{\sqrt{2.3}} \approx 0.66\,.$$

This is the value expected for a coaxial cable with a solid polyethylene dielectric.

From our time-domain reflectometry measurements on the SAT cable (white), we obtained a significantly higher velocity factor of about

$$k_{\text{meas}} \approx 0.79\,.$$

The corresponding effective relative permittivity of the dielectric is

$$\varepsilon_{r,\text{eff}} = \frac{1}{k_{\text{meas}}^2} \approx \frac{1}{0.82^2} \approx 1.5\,.$$

This effective permittivity is much lower than the value for solid polyethylene ($\varepsilon_r \approx 2.3$) and lies in the typical range for foamed (cellular) polyethylene, which is a mixture of polyethylene and air.

Comparing the theoretical velocity factor for solid polyethylene with the measured value, we conclude that the dielectric of the SAT cable is not solid but foamed polyethylene.

5.5.9

For a lossless coaxial cable with inner conductor radius a and inner radius b of the outer conductor, the characteristic impedance is

$$Z_0 = \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{b}{a}\right).$$

Here we have $Z_0 = 75\,\Omega$, an inner conductor diameter of 1 mm (radius $a = 0.5$ mm) and an inner diameter of the outer conductor of 4.3 mm (radius $b = 2.15$ mm), so that

$$\frac{b}{a} = \frac{2.15}{0.5} = 4.3, \qquad 75 = \frac{60}{\sqrt{\varepsilon_r}} \ln(4.3).$$

Solving for ε_r gives

$$\sqrt{\varepsilon_r} = \frac{60\ln(4.3)}{75} \approx \frac{60 \cdot 1.46}{75} \approx \frac{87.5}{75} \approx 1.17, \qquad \varepsilon_r \approx (1.17)^2 \approx 1.4.$$

Thus, the relative permittivity of the dielectric is $\varepsilon_r \approx 1.4$.

5.5.10

For a lossless coaxial cable the characteristic impedance Z_0 is related to the per-unit-length inductance L' and capacitance C' by

$$Z_0 = \sqrt{\frac{L'}{C'}}\,.$$

Solving for L' gives

$$L' = Z_0^2\, C'\,.$$

Using the SAT cable impedance $Z_0 \approx 75\,\Omega$ and the given capacitance per unit length $C' = 56\,\mathrm{pF\,m^{-1}} = 56 \cdot 10^{-12}\,\mathrm{F\,m^{-1}}$, we obtain

$$L' = (75\,\Omega)^2 \cdot 56 \cdot 10^{-12}\,\mathrm{H\,m^{-1}} = 5625 \cdot 56 \cdot 10^{-12}\,\mathrm{H\,m^{-1}} \approx 3.15 \cdot 10^{-7}\,\mathrm{H\,m^{-1}} = 0.315\,\mathrm{\mu H\,m^{-1}}.$$

Thus, the inductance per unit length of the SAT cable is

$$L' \approx 0.32\,\mathrm{\mu H\,m^{-1}}.$$

References

[1] Axel Bangert. *Praktikum Hochfrequenz-Schaltungstechnik*. Kassel.